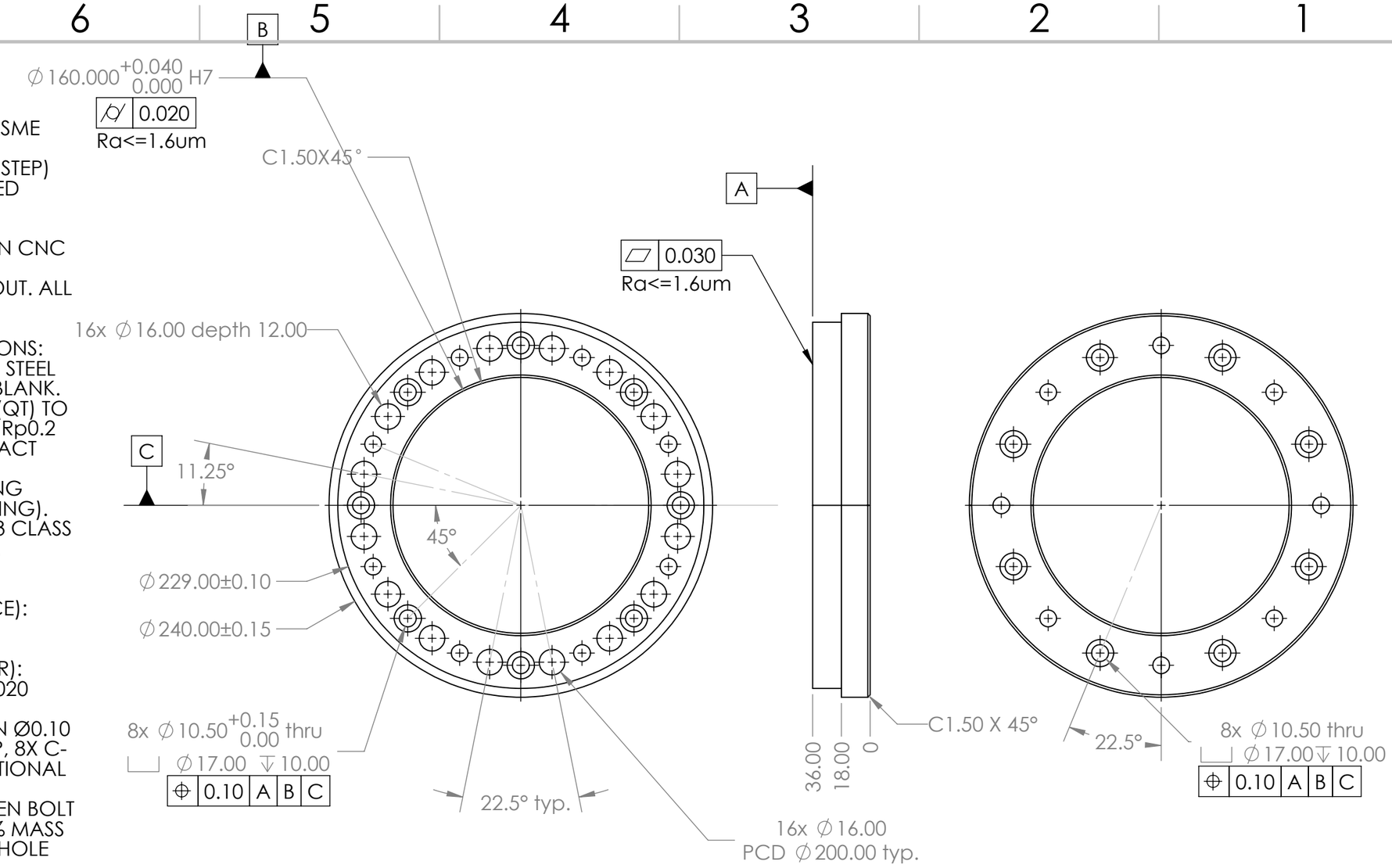
GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

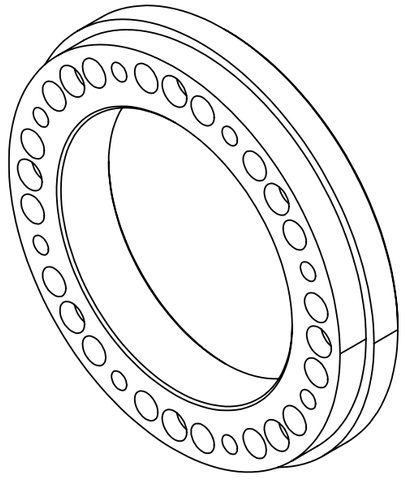
1. GENERAL & STANDARDS COMPLIANCE:
1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET (W3-ENERGY-WIND-RING-6.1.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
1.3 GENERAL TOLERANCES: LINEAR +/- 0.10 mm, ANGULAR +/- 0.2 DEG (HEAVY INDUSTRIAL PRECISION CNC MACHINING).
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.

2. MATERIAL & FATIGUE PERFORMANCE SPECIFICATIONS:
2.1 MATERIAL: HIGH-STRENGTH ALLOY STRUCTURAL STEEL 42CrMo4 (EN 10083-3) OR AISI 4140, FORGED RING BLANK.
2.2 HEAT TREATMENT: QUENCHED AND TEMPERED (QT) TO HARDNESS HB 280-320 (HRC 29-34). YIELD STRENGTH Rp0.2 >= 650 MPa, TENSILE STRENGTH Rm >= 900 MPa, IMPACT ENERGY KV >= 35 J AT -20 DEG C.
2.3 100% HOMOGENEOUS SUBTRACTIVE MACHINING ONLY (CNC VERTICAL TURNING LATHE & 5-AXIS MILLING). 100% ULTRASONIC FLAW INSPECTION PER EN 10228-3 CLASS 3 TO ENSURE ZERO INTERNAL INCLUSIONS OR VOIDS.

3. GEOMETRY & GD&T CONTROLS:
3.1 PRIMARY DATUM [A] (BOTTOM MOUNTING FACE): FLATNESS <= 0.030 mm, SURFACE FINISH Ra <= 1.6 MICRONS.
3.2 SECONDARY DATUM [B] (INNER PILOT DIAMETER): Ø160.00 H7 (+0.040/-0.000 mm), CYLINDRICITY <= 0.020 mm.
3.3 BOLT HOLE MATRIX (16X): TRUE POSITION WITHIN Ø0.10 mm RELATIVE TO [A][B][C]. 8X C-BORES FACING TOP, 8X C-BORES FACING BOTTOM (STAGGERED FOR BI-DIRECTIONAL CLAMPING).
3.4 SKELETAL TRUSS CAVITIES (16X): APPLIED BETWEEN BOLT BOSSES FOR ROTATIONAL WEIGHT REDUCTION (>35% MASS SAVING). MINIMUM WEB WALL THICKNESS BETWEEN HOLE AND POCKET MUST BE >= 3.10 mm.
3.5 OUTER FLANGE SHOULDER: MINIMUM SOLID RADIAL WALL THICKNESS BEHIND BOLT COUNTERBORES MUST BE >= 4.50 mm (NOMINAL 6.00 mm).

4. SURFACE PROTECTION & DEBURRING:
4.1 100% OPTICAL / VISUAL INSPECTION. ALL EDGES BROKEN WITH C1.5 mm CHAMFER OR R >= 2.5 mm TRANSITION. BURR-FREE MANDATORY TO ELIMINATE FATIGUE NOTCH SENSITIVITY.
4.2 COATING SPECIFICATION: ISO 10683 NON-ELECTROLYTICALLY APPLIED ZINC-FLAKE / ZINC-NICKEL COATING (15-20 MICRONS), RESISTANT TO >1500 HOURS NEUTRAL SALT SPRAY TEST (OFFSHORE MARINE C5-M HIGH CORROSION ENVIRONMENT).





For reference

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ±0.10 mm Angular ±0.2°	FINISH: Anti-Corrosion Zinc-Nickel Flake Coating (ISO 10683, C5-M Offshore Marine Grade), Thickness 15-20 µm			DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION: A
	NAME	SIGNATURE	DATE		TITLE: WIND TURBINE BLADE ROOT TRANSITION RING (SKELETAL TRUSS)	
DRAWN						
CHK'D						
APPV'D						
MFG						
Q.A				MATERIAL: High-Strength Alloy Structural Steel 42CrMo4 (EN 10083-3), Quenched & Tempered (HB 280-320)	DWG NO. W3-ENERGY-WIND-RING-6.1	A3
				WEIGHT: ~5.55 kg	SCALE: NTS	SHEET: 1 OF 1